| Execution graph | **Stage 2:** program | **Stage 3:** gem5 harness |
|---|---|---|
| Store bypass<br>i0 `Store` ┐ *rf* *unresolved*<br>i1 `Load` ◄┘ | i0 and i1 share an address; inline asm, so no compiler forwarding. | i0's address held 5000 cycles; i1 forced to predict no alias. |
| Unresolved branch, mispredicted either way<br>i0 `Branch_n` *unresolved* | Constant condition: exits architecturally, so the rest runs only speculatively. | Forced to mispredict; resolution held 5000 cycles; still unresolved at the xmit. |
| Resolved branch, not taken<br>i0 `Branch_n` *resolved* | Zeroed condition: falls through, never taken. | Forced correct; resolution delayed 0, 500 or 2500 cycles (all combinations). |
| Commit boundary<br>i0 lc *committed*<br>i1 fnc *uncommitted* | Global labels on lc and fnc (Alloy instructions only). | Window: lc retired < xmit < fnc retires (measured); no commit stall. |
| Target transmitter<br>i0 `Branch_x` *xmit*<br>*or* `Other_x` *xmit* | `Branch_x`: `testq+jz` on its operand, architecturally taken. Divide: unguarded, so a zero divisor faults. | `Branch_x`: forced to mispredict, no hold. Divide: a fault after a mispredict is a transmission. |